# Cyber Attacks on Systems, Banks and Governments

## Five More Real-World Case Studies (Set 2)

*Central banks, nuclear power, health services, software supply chains and airlines – India and worldwide*

### Overview of the five cases

| # | Case | Date | Region | Reported loss |
|---|---|---|---|---|
| 1 | Bangladesh Bank SWIFT heist | Feb 2016 | Bangladesh / Philippines | $81 million lost; about $15 million recovered from the Philippines leg |
| 2 | Kudankulam nuclear plant malware (DTrack) | Sep–Oct 2019 | India | Administrative network infected; plant systems reported unaffected |
| 3 | SolarWinds supply chain attack | 2020 (found Dec 2020) | USA / worldwide | Fewer than 18,000 customers may have installed tainted software; no single official total cost |
| 4 | Ireland HSE Conti ransomware | May 2021 | Ireland | Forecast cost around €100 million |
| 5 | Air India passenger data breach (SITA) | Feb 2021 | India / worldwide | About 4.5 million passengers' data exposed |

*Note: Figures are as reported by the news articles and reports listed under each case. Where a recovery cost was never officially published, this is stated instead of guessing.*

# Case 1: Bangladesh Bank SWIFT Heist (2016)

| | |
|---|---|
| **Date / timing** | 4–5 February 2016. Malware had been installed in the bank's systems in January 2016. |
| **Victim** | Bangladesh Bank, the central bank of Bangladesh, and its account at the Federal Reserve Bank of New York. |
| **Type of attack** | Custom malware on the bank's SWIFT payment terminals used to send fake payment instructions. |
| **Cause** | Attackers obtained the bank's credentials, planted malware, and studied how international payments were made.<br>The malware sent about three dozen fake SWIFT instructions (nearly $1 billion) and doctored the printed transfer records, so staff did not notice. The attack was timed for the weekend, when Bangladesh and New York offices had different working days. |
| **Loss** | Five requests worth $101 million were paid: $20 million to Sri Lanka and $81 million to accounts at RCBC bank in the Philippines. The New York Fed blocked the remaining requests, worth about $850 million. |
| **Recovery** | The $20 million sent to Sri Lanka was recovered.<br>Most of the $81 million was laundered through Philippine casinos. In 2016 a Philippine court ruled that about $15.25 million belonged to Bangladesh Bank, and it was handed over. |
| **Recovery cost** | Roughly $66 million of the $81 million was still unrecovered according to later reports. No complete figure for investigation and recovery costs has been published. |
| **Solution / lessons** | SWIFT released security updates and tighter controls on member banks.<br>Banks need to separate payment terminals, monitor messages 24x7 and double-check large transfers. |

## What happened

The New York Fed blocked most of the requests after Bangladesh Bank raised the alarm, which stopped most of the stolen amount. The case showed that even central banks can be robbed through their own messaging systems.

## Proof of source (original articles)

- **Wikipedia: Bangladesh Bank robbery –** https://en.wikipedia.org/wiki/Bangladesh_Bank_robbery
- **SecurityWeek: Custom malware used in $81 million Bangladesh Bank heist (BAE Systems findings) –** https://www.securityweek.com/custom-malware-used-81-million-bangladesh-bank-heist/amp/
- **Reuters via EMTV: Bangladesh retrieves a fifth of $81 million stolen –** https://emtv.com.pg/?p=100882
- **ISACA Journal: Lessons learned from the Bangladesh Bank heist –** https://www.isaca.org/resources/isaca-journal/issues/2023/volume-6/lessons-learned-from-the-bangladesh-bank-heist

*Article screenshot / photo:*



# Case 2: Kudankulam Nuclear Power Plant Malware (India, 2019)

| | |
|---|---|
| **Date / timing** | Malware alert from CERT-In on 4 September 2019; became public on 28 October 2019; confirmed by NPCIL on 30 October 2019. |
| **Victim** | Kudankulam Nuclear Power Plant (Tamil Nadu), run by the Nuclear Power Corporation of India Ltd (NPCIL). |
| **Type of attack** | Backdoor malware called DTrack, linked by researchers to the North Korea-linked Lazarus Group. |
| **Cause** | An infected personal computer belonging to a user was connected to the plant's internet-connected administrative network.<br>Security firm Dragos found the malware contained hard-coded IP addresses and a 'KKNPP' username, which pointed to a targeted attack. |
| **Loss** | NPCIL said only the administrative network was affected and that it is isolated from the critical internal network, so reactor control systems were not hit. It was not clear whether data was stolen. |
| **Recovery** | CERT-In informed NPCIL; specialists from the Department of Atomic Energy investigated; the networks were monitored continuously. |
| **Recovery cost** | No cost has been published. |
| **Solution / lessons** | Keep operational networks truly separate from internet-connected ones.<br>Control USB and personal devices; monitor networks continuously. An administrative network can still be a stepping stone to critical systems. |

## What happened

NPCIL first denied the reports and then confirmed them two days later, after a former analyst of India's National Technical Research Organisation posted the finding. The incident raised public concern about cyber security at India's critical infrastructure.

## Proof of source (original articles)

- **POWER Magazine: Malware discovered at nuclear plant in India –** https://www.powermag.com/malware-discovered-at-nuclear-plant-in-india/
- **Dragos: Assessment of reported malware infection at nuclear facility –** https://www.dragos.com/blog/industry-news/assessment-of-reported-malware-infection-at-nuclear-facility
- **Bitdefender: Lazarus Group may have hacked Indian nuclear power plant (includes NPCIL statement) –** https://www.bitdefender.com/en-us/blog/hotforsecurity/lazarus-group-may-have-hacked-indian-nuclear-power-plant
- **The Wire: ISRO, Kudankulam cyber security –** https://cms.thewire.in/economy/tech/isro-kudankulam-cyber-security

*Article screenshot / photo:*



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Appellate Authority

October 30, 2019.

**Press Release**

Identification of malware in NPCIL system is correct. The matter was conveyed by CERT-In when it was noticed by them on September 4, 2019.

The matter was immediately investigated by DAE specialists. The investigation revealed that the infected PC belonged to a user who was connected in the internet connected network used for administrative purposes. This is isolated from the critical internal network. The networks are being continuously monitored.

Investigation also confirms that the plant systems are not affected.

(A.K. Nema) 30/10/19

# Case 3: SolarWinds Supply Chain Attack (2020)

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|---|---|
| **Date / timing** | Tainted updates were sent out between March and June 2020. The attack was announced by security firm FireEye on 13 December 2020. |
| **Victim** | SolarWinds, a US software company, and its customers, including US government agencies (such as Treasury and Commerce) and large companies. |
| **Type of attack** | Supply chain attack: a backdoor called SUNBURST was hidden inside updates of the Orion network management software. |
| **Cause** | Attackers broke into SolarWinds' software build process and added malicious code, so customers installed it as a normal trusted update. The company called it a highly sophisticated attack by an outside nation state. |
| **Loss** | SolarWinds notified about 33,000 Orion customers and said fewer than 18,000 may have installed the affected versions. The hackers could spy on internal email traffic. |
| **Recovery** | Customers had to isolate or rebuild affected systems and reset credentials. MIT Technology Review reported that recovery could take up to 18 months. |
| **Recovery cost** | No single official total cost exists; estimates differ widely, so this document does not quote one. |
| **Solution / lessons** | Verify software updates and secure the build systems of software vendors. Monitor for unusual network activity and limit what management software can access. |

## What happened

The intrusion went unnoticed for months. It was exposed when FireEye found that it had itself been attacked. The case changed how governments and companies think about trusting software updates.

## Proof of source (original articles)

- **Security Affairs: SolarWinds confirms 18,000 customers may have been impacted –** https://securityaffairs.com/?p=112294
- **The Hacker News: Nearly 18,000 SolarWinds customers (SEC filing) –** https://thehackernews.com/2020/12/nearly-18000-solarwinds-customers.html?m=1
- **Cyjax: SolarWinds supply chain attack, summary and analysis –** https://www.cyjax.com/2020/12/21/solarwinds-supply-chain-attack-summary-and-analysis/

*Article screenshot / photo:*



The New York Times

National Edition

SUNDAY, JANUARY 3, 2021

## Scope of Russian Hacking Far Exceeds Initial Fears

*Goal of Intrusion Still Unknown: Was It for Espionage or to Insert 'Back Doors'?*

Ferrying the Infected in California

## Stimulus Puts Deficit Hawks In Lonely Spot

## Covid's Hardships Fray Struggling Communities

"We're Losing Our Grip on Civilization"

# Case 4: Ireland HSE Conti Ransomware (2021)

| | |
|---|---|
| **Date / timing** | Attacker first got in on 18 March 2021 through a phishing email; ransomware was launched on 14 May 2021. |
| **Victim** | Health Service Executive (HSE), Ireland's public health service, with about 54 acute hospitals. |
| **Type of attack** | Conti ransomware (Russia-based group) after weeks of hidden access. |
| **Cause** | A staff member opened a malicious Excel file in a phishing email. The attacker stayed in the network for about eight weeks.<br>An independent PwC report found failings such as antivirus set only to 'monitor' mode, and no central response when early warnings appeared. |
| **Loss** | HSE shut down all its IT systems nationwide. Appointments and services were disrupted for months. The payment system for about 146,000 staff was affected. |
| **Recovery** | Conti demanded $20 million, which HSE refused to pay. The group later gave a decryption key free of charge. Even so, HSE only declared 100% of servers decrypted on 21 September 2021. |
| **Recovery cost** | HSE forecast the overall cost at about €100 million. In June 2021 its director general had said costs could be much higher (reported as over $600 million), so estimates changed over time. |
| **Solution / lessons** | Set antivirus and endpoint tools to block threats, not just monitor.<br>Keep tested offline backups; the PwC report said backups were only periodically copied to tape.<br>Have a central incident response team and plan. |

## What happened

The attack began by a single phishing click. Hospitals first noticed locked computers in the early hours of 14 May, and HSE then shut down everything to contain it. Staff used paper records while systems were rebuilt.

## Proof of source (original articles)

- **Infosecurity Magazine: Ransomware attack could cost Irish health executive €100m –** https://www.infosecurity-magazine.com/news/ransomware-attack-cost-irish
- **Krebs on Security: Inside Ireland's public healthcare ransomware scare –** https://krebsonsecurity.com/2021/12/inside-irelands-public-healthcare-ransomware-scare/
- **HIPAA Journal: HC3 lessons learned from the HSE attack –** https://www.hipaajournal.com/hc3-lessons-learned-from-the-ransomware-attack-on-irelands-health-service-executive/

*Article screenshot / photo:*



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Programmes

# One year on from the HSE cyber attack, could it happen again?

Updated / Saturday, 7 May 2022 07:00

By Brian O'Donovan
Work & Technology Correspondent

At 2.50am on Friday 14 May 2021, the Health Service Executive started to receive the first of multiple reports from hospitals of various IT systems being unavailable. ... executed ransomware attacks on systems within the HSE and across six hospitals.

# Case 5: Air India Passenger Data Breach via SITA (India, 2021)

| | |
|---|---|
| **Date / timing** | SITA was attacked in late February 2021; Air India announced the scale on 21 May 2021. |
| **Victim** | Air India's passengers; the breach was at SITA's Passenger Service System, which also served other airlines such as Singapore Airlines and Lufthansa. |
| **Type of attack** | Data breach through a third-party service provider. |
| **Cause** | Hackers attacked the passenger service system run by SITA, Air India's data processor. Air India said the attack was highly sophisticated. |
| **Loss** | Personal data of about 4.5 million passengers worldwide, registered between August 2011 and February 2021: names, dates of birth, contact, passport, ticket, frequent-flyer and credit card details. Air India said no password data was breached. |
| **Recovery** | Air India said the compromised servers were secured and no abnormal activity was seen afterwards. It informed regulators in India and abroad and asked passengers to change passwords. |
| **Recovery cost** | No cost figure has been published. |
| **Solution / lessons** | Check the security of third-party vendors that hold customer data.<br>Notify affected people quickly; Air India said it received the identities of affected passengers only on 25 March and 5 April 2021. |

## What happened

The breach did not happen inside Air India's own network but at its vendor, which shows that a company's data is only as safe as its suppliers' systems.

## Proof of source (original articles)

- **AP via Click2Houston: India's national carrier says hack leaked passengers' data –** https://www.click2houston.com/business/2021/05/23/indias-national-carrier-says-hack-leaked-data-of-passengers
- **FinTech Global: Air India reveals data of 4.5 million passengers was stolen –** https://fintech.global/2021/05/25/air-india-reveals-data-of-4-5-million-passengers-was-stolen-in-sita-cyberattack/
- **Security Affairs: Air India suffered a data breach –** https://securityaffairs.com/118162/data-breach/air-india-data-breach.html
- **Khaleej Times: Air India passengers' data leaked –** https://www.khaleejtimes.com/business/aviation/air-india-passengers-data-leaked-after-cyberattack-on-its-service-system-provider?amp=1

*Article screenshot / photo:*



## Conclusion: What these cases show

- Attackers often wait weeks or months before striking (Bangladesh Bank, HSE, SolarWinds).
- Trusted channels can be abused: payment messaging, software updates and third-party vendors.
- A single infected device or phishing click can open a network (Kudankulam, HSE).
- Delayed notification and weak monitoring make losses bigger (Air India, Bangladesh Bank).

Common defences: staff training against phishing, strong monitoring that blocks threats, strict separation of critical networks, vendor security checks, tested offline backups and a ready incident response plan.

### Important note for submission

The dashed boxes in each case are for you to paste a screenshot of the original article. Open a link under 'Proof of source', take a screenshot of the headline and date, and paste it in. Keep the links as well.